

Molecular Epidemiologic Evidence for Diabetogenic Effects of Dioxin Exposure in U.S. Air Force Veterans of the Vietnam War

Background One of the outcomes positively associated with dioxin exposure in humans is type 2 diabetes. **Objectives** This study was conducted in order to find the molecular biological evidence for the diabetogenic action of dioxin in adipose samples from Vietnam veterans. **Methods** We obtained 313 adipose tissue samples both from Vietnam veterans who were exposed to dioxin (Operation Ranch Hand) and from comparison veterans who served in Southeast Asia with no record of dioxin exposure. We conducted quantitative reverse-transcribed polymerase chain reaction studies on selected marker mRNAs from these samples. **Results** We found the most sensitive and reliable molecular indicator of dioxin-induced diabetes to be the ratio of mRNA of glucose transporter 4 (GLUT4) and nuclear transcription factor kappa B (NFκB), a marker of inflammation. This ratio showed significant correlations to serum dioxin residues and to fasting glucose among those in the Ranch Hand group and, surprisingly, even in the comparison group, who have low levels of dioxin comparable to the general public. Such a correlation in the comparison group was particularly significant among those with known risk factors such as obesity and family history of diabetes. **Conclusions** These results show that the GLUT4:NFκB ratio is a reliable marker for the diabetogenic action of dioxin, particularly at very low exposure levels that are not much higher than those found in the general public, implying a need to address current exposure levels.